Mathematics: analysis and approaches
Standard level
Paper 2
2.1–2.5 Functions

1 November 2026

W4ss Academy practice set 1 | **Topic** 2.1–2.5 Functions | **Ref.** W4SS–MAA–T2–P2–01

Student name

30 minutes

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- A graphic display calculator is required for this paper.
- This paper has a Section A only. Answer all questions. Answers must be written within the answer boxes provided.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[25 marks]**.

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

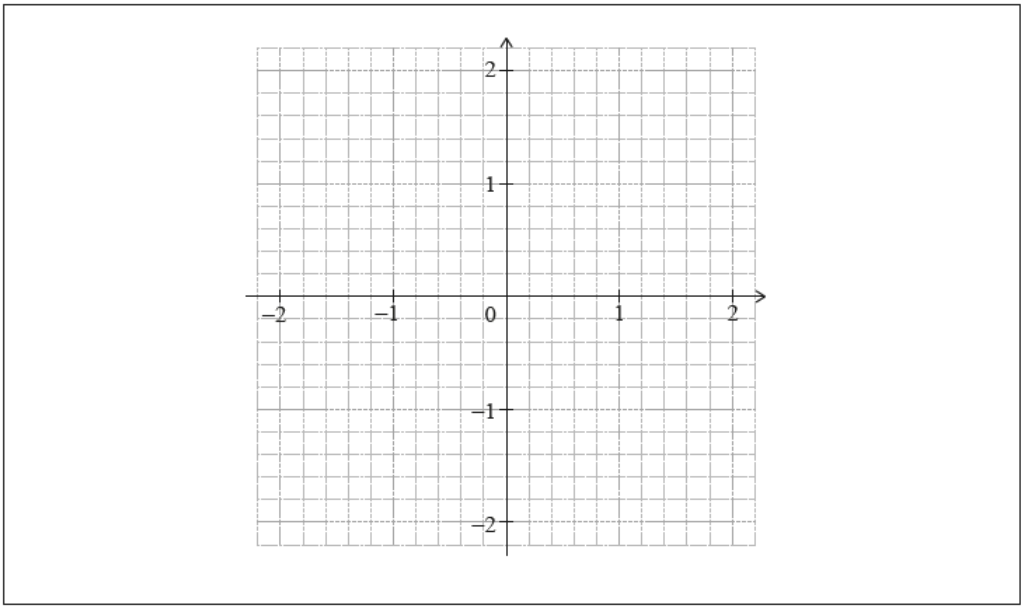
Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 5]

Consider the function $f(x) = e^{-x^2} - 0.5$, for $-2 \leq x \leq 2$.

(a) Find the values of x for which $f(x) = 0$. [2]

(b) Sketch the graph of f on the following grid.



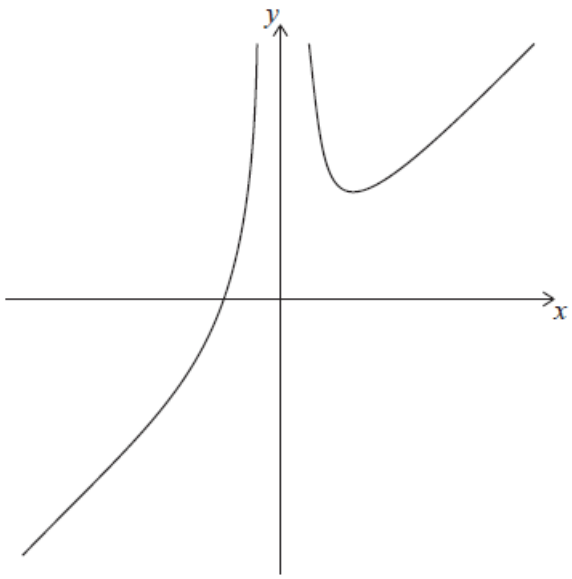
[3]

(This question continues on the following page)

(Question 1 continued)

2. [Maximum mark: 6]

Consider the graph of the function $f(x) = x + \frac{12}{x^2}$, $x \neq 0$.



(a) (i) Write down the zero of $f(x)$. [2]

(ii) Write down the coordinates of the local minimum point. [2]

(b) Consider the function $g(x) = 3 - x$.

Solve $f(x) = g(x)$. [2]

(Question 2 continued)

3. [Maximum mark: 6]

The functions f and g are defined for $x \in \mathbb{R}$ by $f(x) = 6x^2 - 12x + 1$ and $g(x) = -x + c$, where $c \in \mathbb{R}$.

(a) Find the range of f. [2]

(b) Given that $(g \circ f)(x) \leq 0$ for all $x \in \mathbb{R}$, determine the set of possible values for c. [4]

4. [Maximum mark: 8]

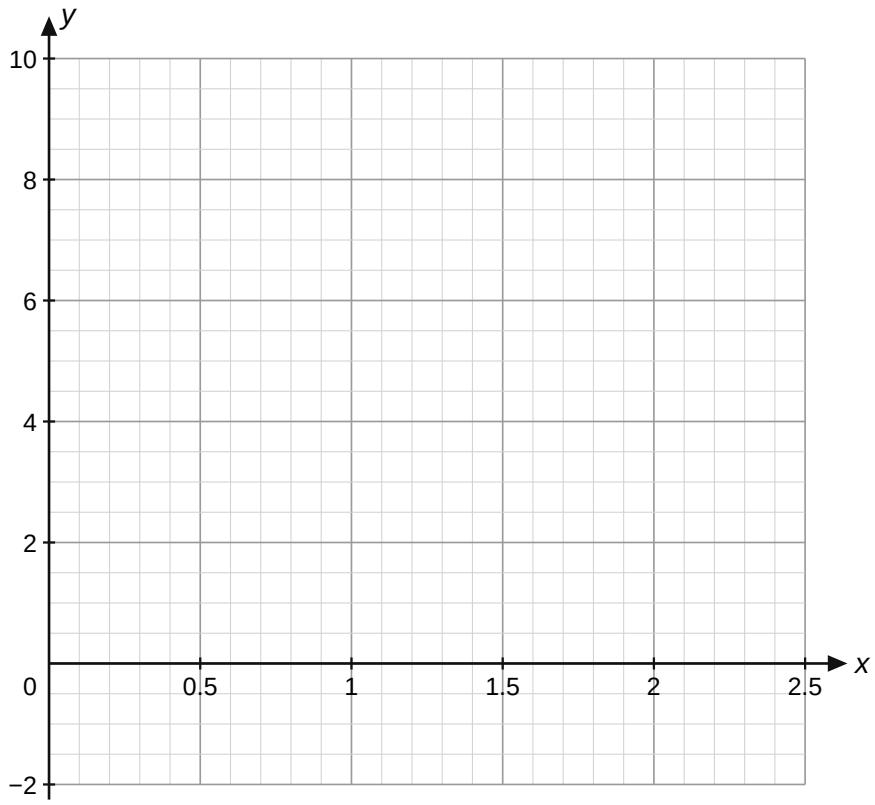
Let $f(x) = x^2 - 1$ and $g(x) = x^2 - 2$, for $x \in R$.

(a) Show that $(f \circ g)(x) = x^4 - 4x^2 + 3$. [2]

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(Question 4 continued)

(b) On the following grid, sketch the graph of $(f \circ g)(x)$, for $0 \leqslant x \leqslant 2.25$.



[3]

(c) The equation $(f \circ g)(x) = k$ has exactly two solutions, for $0 \leqslant x \leqslant 2.25$. Find the possible values of k. [3]

(Question 4 continued)

Sources of the questions

This practice paper was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Mathematics: analysis and approaches Questionbank (standard level, topic 2 Functions) and cover subtopics 2.1 to 2.5. Some of them come from former courses (Mathematics SL and HL, Mathematical studies SL). They remain the property of the International Baccalaureate Organization and must not be published or sold.

Section A — question 1	21N.2.SL.TZ0.2 · 2.3 sketching a graph · 2.4 zeros
Section A — question 2	20N.1.SL.TZ0.T_4 (former Mathematical studies SL course) · 2.4 zero, local minimum, intersection using technology — Mathematical studies Paper 1, where a calculator was allowed
Section A — question 3	21M.2.SL.TZ2.5 · 2.2 range · 2.5 composite function
Section A — question 4	17M.2.SL.TZ2.S_6 (former Mathematics SL course) · 2.5 composite function · 2.3 sketching · 2.4 number of solutions — figure redrawn by W4ss Academy from the data in the question

Please **do not** write on this page.

Answers written on this page
will not be marked.